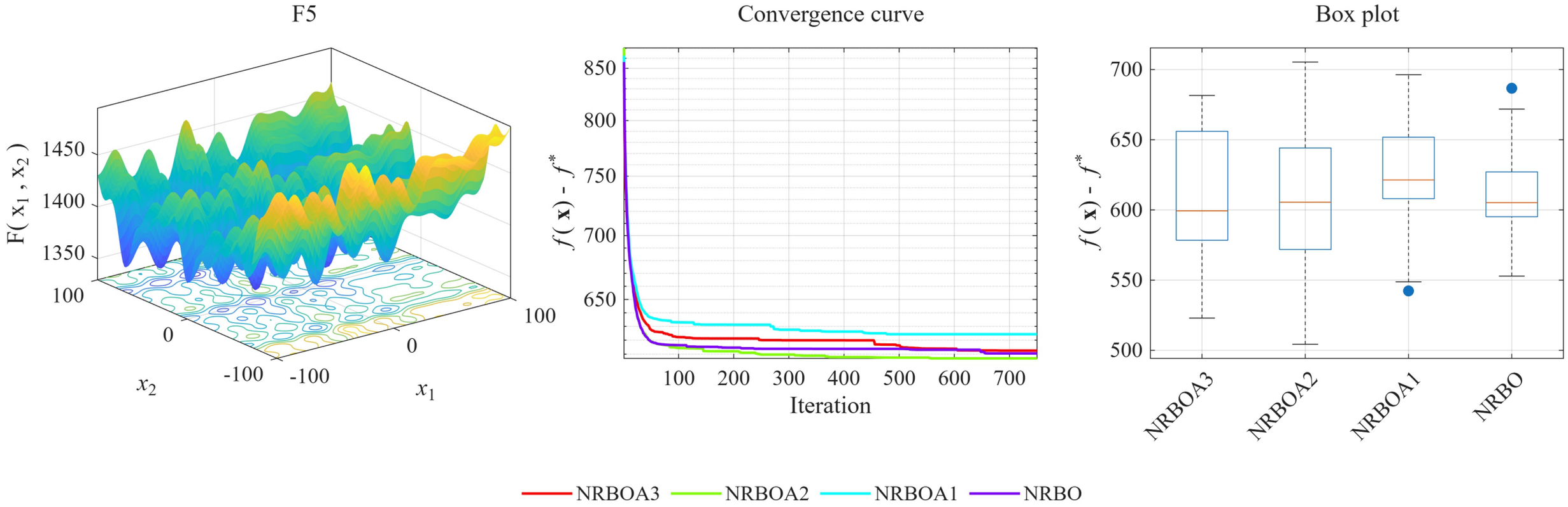

F5
F( x1 , x2 )
1450
1400
1350
100
0
-100
x2
-100
0
100
x1
Convergence curve
f( x) - f*
850
800
750
700
650
100
200
300
400
500
600
700
Iteration
Box plot
f( x) - f*
700
650
600
550
500
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO